Stochastic Processes — 1060429

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Throughout, a Brownian motion (BM) is standard ($B_0 = 0$) with continuous paths, and $\mathcal{F}_t$ denotes the filtration in use. Apparent typos are pointed out where they occur.

Exercise 1

Relations between stochastic processes

1. Let $\Omega_0 = \{X, Y \text{ right-continuous}\} \cap \bigcap_{q\in\mathbb{Q}_+}\{X_q = Y_q\}$. The first event contains a measurable set of probability 1, and each $\{X_q = Y_q\}$ has probability 1 (modification), so Ω_0 contains a measurable set of probability 1 (countable intersection). For $\omega \in \Omega_0$ and $t \geq 0$ take $q_n \in \mathbb{Q}_+$, $q_n \downarrow t$; then $X_t(\omega) = \lim X_{q_n}(\omega) = \lim Y_{q_n}(\omega) = Y_t(\omega)$. Hence $X_t = Y_t$ for all t on Ω_0, i.e. X, Y are indistinguishable.

2. Let $Q_0 = \mathbb{Q} \cap [0, t_0)$ and

$$C_{t_0} = \bigcap_{k\geq 1}\bigcup_{m\geq 1}\bigcap_{s,u\in Q_0,\ |s-u|<1/m} \{|X_s - X_u| \leq 1/k\} \in \sigma(X_t : t \leq t_0).$$

Claim. If the path $X(\omega)$ is RCLL on $[0, t_0]$, then $\omega \in A_{t_0} \iff \omega \in C_{t_0}$.

If $X(\omega)$ is continuous on $[0, t_0)$ and has a left limit at t_0, it extends continuously to $[0, t_0]$, so it is uniformly continuous on $[0, t_0)$ and $\omega \in C_{t_0}$. Conversely, let $\omega \in C_{t_0}$, $k \geq 1$, and m as in the definition. For $s, u \in [0, t_0)$ with $|s - u| < 1/m$ choose $q_j, r_j \in Q_0$ with $q_j \downarrow s$, $r_j \downarrow u$ and $|q_j - r_j| < 1/m$ (possible as $s, u < t_0$). By right-continuity $|X_s - X_u| = \lim |X_{q_j} - X_{r_j}| \leq 1/k$. So $X(\omega)$ is uniformly continuous on $[0, t_0)$, hence $\omega \in A_{t_0}$.

(a) If every path is RCLL, the claim gives $A_{t_0} = C_{t_0} \in \sigma(X_t : t \leq t_0)$.

(b) Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, $\mathbb{P} = \text{Leb}$, and let C be the Cantor set. Since $|\mathcal{P}(C)| = 2^{\mathfrak{c}} > \mathfrak{c} = |\mathcal{B}([0, 1])|$, there is $E \subset C$ with $E \notin \mathcal{F}$. Define $X_t(\omega) = \mathbf{1}_E(\omega)\mathbf{1}_{\{t=\omega\}}$. Then $X_t = \mathbf{1}_{E\cap\{t\}}$ is the indicator of the empty set or of a single point, so it is measurable. For $\omega \notin E$ the path is $\equiv 0$. For $\omega \in E$ the path has a spike at $t = \omega$ and is not right-continuous there. Thus $\{X \text{ RCLL}\} = \Omega \setminus E \supseteq \Omega \setminus C$ and $\mathbb{P}(\Omega \setminus C) = 1$, so X is RCLL with probability 1. But $A_2 = \Omega \setminus E \notin \mathcal{F}$, and in particular $A_2 \notin \sigma(X_t : t \leq 2) \subseteq \mathcal{F}$.

(c) There is $N \in \mathcal{F}$ with $\mathbb{P}(N) = 0$ such that every path outside N is RCLL. By the claim, $A_{t_0} \triangle C_{t_0} \subseteq N$. Completeness of $\mathcal{F}_{t_0}$ (it contains every subset of a $\mathbb{P}$-null set) gives $A_{t_0} \setminus C_{t_0}$, $C_{t_0} \setminus A_{t_0} \in \mathcal{F}_{t_0}$. Hence $A_{t_0} = (C_{t_0} \setminus (C_{t_0} \setminus A_{t_0})) \cup (A_{t_0} \setminus C_{t_0}) \in \mathcal{F}_{t_0}$.

Brownian motion

1.

Remark. The condition should read $\mathbb{E}[|X_t - X_s|^\alpha] \leq M|t - s|^{1+\beta}$ for all $t, s \in [0, T]$. In (c) the bound should be $2^{-\gamma n}$, not 2^{-m}. In (e),(f) the index set is $[0, 1]$.

(a) Put $\tilde{X}_u = X_{uT}$, $u \in [0, 1]$. Then $\mathbb{E}|\tilde{X}_u - \tilde{X}_v|^\alpha \leq MT^{1+\beta}|u - v|^{1+\beta}$, which is the hypothesis on $[0, 1]$ with constant $MT^{1+\beta}$. If $\tilde{Y}$ is a modification of $\tilde{X}$ that is a.s. γ-Hölder, then $Y_t = \tilde{Y}_{t/T}$ satisfies $\mathbb{P}(Y_t = X_t) = \mathbb{P}(\tilde{Y}_{t/T} = \tilde{X}_{t/T}) = 1$, and $|Y_t - Y_s| \leq K|t/T - s/T|^\gamma = KT^{-\gamma}|t - s|^\gamma$.

(b) By Markov's inequality, $\mathbb{P}(|X_t - X_s| > \varepsilon) \leq \varepsilon^{-\alpha} M|t - s|^{1+\beta} \to 0$ as $t \to s$.

(c) By a union bound and Markov's inequality,

$$\mathbb{P}\Big(\max_{1\leq k\leq 2^n} |X_{k2^{-n}} - X_{(k-1)2^{-n}}| > 2^{-\gamma n}\Big) \leq 2^n \cdot 2^{\gamma\alpha n} M 2^{-n(1+\beta)} = M 2^{-n(\beta-\alpha\gamma)},$$

which is summable since $\gamma < \beta/\alpha$. By Borel–Cantelli, a.s. only finitely many of these events occur. Let $N(\omega)$ be the last n at which one occurs ($N = 0$ if none occurs). Then $\mathbb{P}(N < \infty) = 1$ and $\max_k |X_{k2^{-n}} - X_{(k-1)2^{-n}}| \le 2^{-\gamma n}$ for all $n > N(\omega)$.

(d) Fix ω with $N(\omega) < \infty$. *Claim:* for $m > n \ge N(\omega)$ and $s < t$ in D_m with $t - s < 2^{-n}$,

$$|X_t - X_s| \le 2 \sum_{j=n+1}^{m} 2^{-\gamma j}.$$

We use induction on m. For $m = n + 1$ we have $t - s = 2^{-(n+1)}$, and the bound holds by (c). For the step $m \to m + 1$, let $s, t \in D_{m+1}$. Let $s' = \min\{r \in D_m : r \ge s\}$ and $t' = \max\{r \in D_m : r \le t\}$. Then $s' - s,\ t - t' \in \{0, 2^{-(m+1)}\}$, and $s' \le t'$ (otherwise $s < t < s' \le s + 2^{-(m+1)}$, which is impossible in D_{m+1}). Since $t' - s' \le t - s < 2^{-n}$, the induction hypothesis (or $t' = s'$) and (c) at level $m + 1 > N$ give

$$|X_t - X_s| \le 2 \sum_{j=n+1}^{m} 2^{-\gamma j} + 2 \cdot 2^{-\gamma(m+1)}.$$

Now let $s < t$ in $D = \bigcup_m D_m$ with $t - s < 2^{-N(\omega)}$. Choose n with $2^{-(n+1)} \le t - s < 2^{-n}$, so $n \ge N(\omega)$, and choose $m > n$ with $s, t \in D_m$. Then

$$|X_t - X_s| \le 2 \sum_{j>n} 2^{-\gamma j} = \frac{2 \cdot 2^{-\gamma(n+1)}}{1 - 2^{-\gamma}} \le C_\gamma |t - s|^\gamma, \qquad C_\gamma = \frac{2}{1 - 2^{-\gamma}}.$$

(e) Let $s < t$ in D with $t - s \ge 2^{-N}$. The points of $D_{N+1} \cap (s, t)$ together with s, t form a chain $s = u_0 < \cdots < u_k = t$ in D with consecutive gaps $< 2^{-N}$ and $k \le 2^{N+1} + 1$. Each gap contributes at most $C_\gamma 2^{-N\gamma} \le C_\gamma |t - s|^\gamma$. Hence on $\{N < \infty\}$,

$$|X_t - X_s| \le K(\omega)|t - s|^\gamma \text{ for all } s, t \in D, \qquad K = (2^{N+1} + 1)C_\gamma.$$

(f) On $\Omega^* = \{N < \infty\}$, $X|_D$ is uniformly continuous, so $Y_t := \lim_{D \ni r \to t} X_r$ exists for all $t \in [0, 1]$. Set $Y \equiv 0$ off Ω^*. Each Y_t is measurable (a limit along a sequence), and Y is γ-Hölder on $[0, 1]$ with constant K by continuity. For fixed t take $r_j \in D$, $r_j \to t$: then $X_{r_j} \to Y_t$ a.s. and $X_{r_j} \to X_t$ in probability by (b). Limits in probability are a.s. unique, so $Y_t = X_t$ a.s., and Y is a modification of X. Y does not depend on γ. Taking $\gamma_k \uparrow \beta/\alpha$ and intersecting the corresponding events $\Omega^*_{\gamma_k}$ shows that a.s. Y is γ-Hölder for every $\gamma < \beta/\alpha$.

2. Here t_0 is a local maximum if there is $\varepsilon > 0$ with $B_s \le B_{t_0}$ for all $s \in [0, \infty) \cap (t_0 - \varepsilon, t_0 + \varepsilon)$.

(a) $W_s = B_{t_0+s} - B_{t_0}$ is a BM. If t_0 is a local maximum, then $\max_{[0,1/n]} W \le 0$ for some n. By the reflection principle,

$$\mathbb{P}\big(\max_{[0,1/n]} W \ge a\big) = 2\mathbb{P}(W_{1/n} \ge a) \xrightarrow[a \downarrow 0]{} 1,$$

so $\mathbb{P}(\max_{[0,1/n]} W > 0) = 1$ for every n. Hence $\mathbb{P}\big(\bigcup_n \{\max_{[0,1/n]} W \le 0\}\big) = 0$, and $\{t_0 \text{ local max}\}$ is contained in a null set.

(b) B attains its maximum on $[0, 1]$ at some t^*. By (a) with $t_0 = 0$, a.s. $t^* \ne 0$. The process $W_s = B_1 - B_{1-s}$, $s \in [0, 1]$, is a continuous centred Gaussian process. Its covariance is

$$\operatorname{Cov}(W_s, W_u) = 1 - (1 - u) - (1 - s) + (1 - s \vee u) = s \wedge u,$$

so W is a BM on $[0, 1]$. Moreover $t^* = 1$ means $\min_{[0,1]} W \ge 0$, which has probability 0 by the argument of (a) applied to $-W$. So a.s. $t^* \in (0, 1)$, and t^* is a local maximum of B.

3.

(a) Let $Z_n = (B_{1/n} - B_{1/(n+1)})/\sqrt{1/n - 1/(n+1)}$. The intervals $[\frac{1}{n+1}, \frac{1}{n}]$ are disjoint, so the Z_n are i.i.d. $N(0,1)$. Then $\sum_n \mathbb{P}(Z_n \geq K) = \sum_n \mathbb{P}(Z_1 \geq K) = \infty$, and the second Borel–Cantelli lemma gives $\mathbb{P}(Z_n \geq K \text{ i.o.}) = 1$.

(b) Fix rationals $p < q$. The process $W_t = B_{p+t} - B_p$ is a BM. Applying (a) to W for every $K \in \mathbb{N}$, a.s. for all K there are infinitely many n with $1/n \leq q - p$ and

$$|W_{1/n} - W_{1/(n+1)}| \geq K|1/n - 1/(n+1)|^{1/2}.$$

So B is not $\frac{1}{2}$-Hölder on $[p,q]$. Intersecting over the countably many pairs (p,q), and using that every non-trivial interval contains some $[p,q]$, gives the claim.

4. Let $n \in \mathbb{N}$, $h = 1/n$, $t_k = kh$, $\delta = \varepsilon/4$. Let L be the piecewise linear interpolation of $(t_k, B_{t_k})_k$ and $Z = B - L$. On $[t_{k-1}, t_k]$,

$$Z_s = (B_s - B_{t_{k-1}}) - \tfrac{s-t_{k-1}}{h}(B_{t_k} - B_{t_{k-1}}), \qquad \text{so} \quad \sup_{[t_{k-1},t_k]} |Z| \leq 2 \sup_{u\in[0,h]} |B_{t_{k-1}+u} - B_{t_{k-1}}|.$$

Independence. $(Z_s)_s$ and $(B_{t_j})_j$ are jointly Gaussian and $\operatorname{Cov}(Z_s, B_{t_j}) = 0$. For $t_j \leq t_{k-1}$ this covariance equals $t_j - \frac{(t_k-s)t_j + (s-t_{k-1})t_j}{h} = 0$, and for $t_j \geq t_k$ it equals $s - \frac{(t_k-s)t_{k-1} + (s-t_{k-1})t_k}{h} = 0$. Hence $\sigma(Z_s : s \in [0,1])$ is independent of $(B_{t_1}, \dots, B_{t_n})$.

Let f_h be the linear interpolation of $(t_k, f(t_k))$. Then $|f_h - f| \leq \omega_f(h)$, where ω_f is the modulus of continuity. Consider the events

$$A = \{\max_k |B_{t_k} - f(t_k)| < \delta\}, \qquad C = \{\sup_{[0,1]} |Z| < \delta\}.$$

On $A \cap C$ we have $|B_t - f(t)| \leq |L_t - f_h(t)| + |f_h(t) - f(t)| + |Z_t| < 2\delta + \omega_f(h)$, which is $< \varepsilon$ once $\omega_f(h) < \varepsilon/2$.

$\mathbb{P}(A) > 0$, since $(B_{t_1}, \dots, B_{t_n})$ is an invertible linear image of i.i.d. $N(0,h)$ increments and so has an everywhere positive density. The suprema over the different intervals are independent (independent increments) and each has the law of $\sup_{[0,h]} |B|$. By scaling,

$$\mathbb{P}(C) \geq \mathbb{P}\big(\sup_{[0,h]} |B| < \delta/2\big)^n = \mathbb{P}\big(\sup_{[0,1]} |B| < \delta/(2\sqrt{h})\big)^n.$$

The inner probability tends to 1 as $h \to 0$, so it is positive for h small. Choosing such h with also $\omega_f(h) < \varepsilon/2$, independence gives $\mathbb{P}(\sup_{[0,1]} |B - f| < \varepsilon) \geq \mathbb{P}(A)\mathbb{P}(C) > 0$.

5.

(a) $X_0 = 0$ and X has continuous paths. For $0 \leq t_0 < \cdots < t_k$, the increments $X_{t_i} - X_{t_{i-1}} = a \cdot (B_{t_i} - B_{t_{i-1}})$ are functions of the independent vectors $B_{t_i} - B_{t_{i-1}}$, hence independent. Each is a sum of independent $N(0, a_j^2(t_i - t_{i-1}))$ variables, i.e. $N(0, |a|^2(t_i - t_{i-1})) = N(0, t_i - t_{i-1})$. So X is a BM.

(b) By (a) each coordinate $Y^i = a_i \cdot B$ is a BM, where a_i is the i-th row. All Y_t^i are linear combinations of the Gaussian family (B_t^j), so $(Y^1, \dots, Y^m)$ is jointly Gaussian with $\operatorname{Cov}(Y_s^i, Y_t^j) = (a_i \cdot a_j)(s \wedge t) = \delta_{ij}(s \wedge t)$. Uncorrelated jointly Gaussian families are independent: the finite-dimensional distributions factor, and a π–λ argument finishes the proof. So $Y^1, \dots, Y^m$ are independent BMs.

(c) For $R \in SO(d)$ the rows are orthonormal, so by (b) RB consists of d independent BMs, i.e. it is a d-dimensional BM. Since the law of a process is determined by its finite-dimensional distributions, $(RB_t)_t \overset{d}{=} (B_t)_t$.

6. For a partition π of $[0,T]$ write $Q_\pi(X) = \sum_i (X_{t_i} - X_{t_{i-1}})^2$. We use the known result that $Q_\pi(B) \to b - a$ in L^2 on any interval $[a,b]$; indeed $\mathbb{E}(Q_\pi(B) - (b-a))^2 = 2\sum \Delta t_i^2 \leq 2(b-a)|\pi|$.

(a) $Q_\pi(aB) = a^2 Q_\pi(B) \to a^2 T$.

(b) $W_t = c^{-1/2} B_{ct}$ is a BM, so $Q_\pi(B_{c\cdot}) = c Q_\pi(W) \to cT$.

(c) Let $Y_t = \int_0^t B_s\, ds$ and $M = \sup_{[0,T]} |B|$. Then $|Y_{t_i} - Y_{t_{i-1}}| \le M \Delta t_i$, so $Q_\pi(Y) \le M^2 T |\pi|$ and $\mathbb{E} Q_\pi(Y)^2 \le \mathbb{E}[M^4] T^2 |\pi|^2 \to 0$. Here $\mathbb{E} M^4 < \infty$, since $M \le \sup B + \sup(-B)$ and $\sup_{[0,T]} B \overset{d}{=} |B_T|$ by the reflection principle. The quadratic variation is 0.

(d) If $t_0 \in \pi$, then $Q_\pi(X) = Q_{\pi \cap [0,t_0]}(B) + 9\, Q_{\pi \cap [t_0,T]}(B) \to t_0 + 9(T - t_0)$ in L^2. Otherwise let $[t_{j-1}, t_j] \ni t_0$, $u = B_{t_0} - B_{t_{j-1}}$, $v = -3(B_{t_j} - B_{t_0})$. Then $Q_\pi(X) = Q_{\pi \cup \{t_0\}}(X) + 2uv$, and by independence $\mathbb{E}(2uv)^2 = 36(t_0 - t_{j-1})(t_j - t_0) \le 36|\pi|^2 \to 0$. Hence the quadratic variation is $t_0 + 9(T - t_0) = 9T - 8t_0$.

Exercise 2

Conditional expectation

1.

(a) Let $Z = \mathbb{E}[X|Y]$. By conditional Jensen, $Z^2 \le \mathbb{E}[X^2|Y]$, so $Z \in L^2$. If $\mathbb{E}[h(Y)^2] = \infty$ the right-hand side is infinite (as $X \in L^2$). Otherwise $V = Z - h(Y) \in L^2(\sigma(Y))$, and by Problem 2(b) (the product $V(X - Z)$ is integrable by Cauchy–Schwarz)

$$\mathbb{E}[V(X - Z)] = \mathbb{E}\big[V\, \mathbb{E}[X - Z \,|\, Y]\big] = 0.$$

Hence $\mathbb{E}(X - h(Y))^2 = \mathbb{E}(X - Z)^2 + \mathbb{E}V^2 + 2\mathbb{E}[V(X - Z)] \ge \mathbb{E}(X - Z)^2$.

(b) $L^2(\Omega, \sigma(Y), \mathbb{P})$ is a closed subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$. As shown, $Z = \mathbb{E}[X|Y]$ lies in it, and $\mathbb{E}[V(X - Z)] = 0$ for every $V \in L^2(\sigma(Y))$, i.e. $X - Z \perp L^2(\sigma(Y))$. These two properties characterise the orthogonal projection. (Equivalently, by (a): every $\sigma(Y)$-measurable V is of the form $h(Y)$, so Z is the closest point.)

2.

(a) ($\Rightarrow$) Let $Z = \mathbb{E}[X|\mathcal{G}]$. For $W = \mathbf{1}_A$, $A \in \mathcal{G}$, the identity is the definition. By linearity it holds for simple W. For W with $|W| \le c$ a.s., take simple $\mathcal{G}$-measurable $W_n \to W$ with $|W_n| \le c$. Then $|W_n Z| \le c|Z|$ and $|W_n X| \le c|X|$, and dominated convergence gives $\mathbb{E}[WZ] = \mathbb{E}[WX]$.

($\Leftarrow$) Taking $W = \mathbf{1}_A$ gives $\mathbb{E}[Z\mathbf{1}_A] = \mathbb{E}[X\mathbf{1}_A]$ for $A \in \mathcal{G}$. Integrability of Z: with $W_n = \operatorname{sgn}(Z)\mathbf{1}_{\{|Z| \le n\}}$ we get $\mathbb{E}[|Z|\mathbf{1}_{\{|Z| \le n\}}] = \mathbb{E}[W_n X] \le \mathbb{E}|X|$. Letting $n \to \infty$ gives $\mathbb{E}|Z| \le \mathbb{E}|X|$ by monotone convergence. Hence $Z = \mathbb{E}[X|\mathcal{G}]$.

(b) First let $X, Y \ge 0$ and $Z = \mathbb{E}[X|\mathcal{G}] \ge 0$. For $A \in \mathcal{G}$, $W = \mathbf{1}_A \min(Y, n)$ is bounded and $\mathcal{G}$-measurable, so $\mathbb{E}[\mathbf{1}_A \min(Y, n) Z] = \mathbb{E}[\mathbf{1}_A \min(Y, n) X]$. By monotone convergence, $\mathbb{E}[\mathbf{1}_A YZ] = \mathbb{E}[\mathbf{1}_A YX] < \infty$. So YZ is integrable, $\mathcal{G}$-measurable, and equals $\mathbb{E}[XY|\mathcal{G}]$.

In general write $X = X^+ - X^-$ and $Y = Y^+ - Y^-$. Each $X^\pm Y^\pm \le |XY|$ is integrable, and the positive case gives $\mathbb{E}[X^\pm Y^\pm|\mathcal{G}] = Y^\pm \mathbb{E}[X^\pm|\mathcal{G}]$. Summing with signs and using linearity,

$$\mathbb{E}[XY|\mathcal{G}] = (Y^+ - Y^-)(\mathbb{E}[X^+|\mathcal{G}] - \mathbb{E}[X^-|\mathcal{G}]) = Y\, \mathbb{E}[X|\mathcal{G}].$$

3.

$\mathcal{G}$ is generated by the partition $I_1 = [0, \frac{1}{4}]$, $I_2 = (\frac{1}{4}, \frac{2}{3}]$, $I_3 = (\frac{2}{3}, 1]$, so $\mathcal{G} = \{\bigcup_{i \in S} I_i : S \subseteq \{1, 2, 3\}\}$. Let $Z = \sum_i a_i \mathbf{1}_{I_i}$ with $a_i = \mathrm{Leb}(I_i)^{-1} \int_{I_i} \omega^2 d\omega$. Then Z is $\mathcal{G}$-measurable, and $\mathbb{E}[Z\mathbf{1}_{I_i}] = \mathbb{E}[X\mathbf{1}_{I_i}]$ for each i, hence for every $A \in \mathcal{G}$ by additivity. By uniqueness $\mathbb{E}[X|\mathcal{G}] = Z$ a.s. The coefficients are

$$a_1 = 4 \cdot \tfrac{1}{3 \cdot 64} = \tfrac{1}{48}, \qquad a_2 = \tfrac{12}{5} \cdot \tfrac{1}{3}\Big(\tfrac{8}{27} - \tfrac{1}{64}\Big) = \tfrac{12}{5} \cdot \tfrac{485}{5184} = \tfrac{97}{432}, \qquad a_3 = 3 \cdot \tfrac{1}{3}\Big(1 - \tfrac{8}{27}\Big) = \tfrac{19}{27}.$$

4.

Remark. Y must be allowed to take the value $+\infty$. For instance, if $\mathcal{G} = \{\emptyset, \Omega\}$ then $\int_\Omega Y\, d\mathbb{P} = \mathbb{E}X = \infty$ forces $Y \equiv \infty$. We prove the statement for $[0,\infty]$-valued Y.

Existence. For $M \in \mathbb{N}$ let $X_M = \min(X, M)$ and $Y_M = \mathbb{E}[X_M|\mathcal{G}]$. Since $0 \le X_M \le X_{M+1}$, monotonicity gives $0 \le Y_M \le Y_{M+1}$ a.s. Outside a null set the sequence is non-decreasing, and $Y := \lim_M Y_M \in [0,\infty]$ is $\mathcal{G}$-measurable. For $A \in \mathcal{G}$, monotone convergence gives

$$\int_A Y = \lim_M \int_A Y_M = \lim_M \int_A X_M = \int_A X.$$

Uniqueness. Let Y, Y' be two such variables and $A_n = \{Y' > Y,\ Y \le n\} \in \mathcal{G}$. Then $\int_{A_n} Y' = \int_{A_n} X = \int_{A_n} Y \le n < \infty$, so $\int_{A_n}(Y' - Y) = 0$ with $Y' - Y > 0$ on A_n. Hence $\mathbb{P}(A_n) = 0$, and taking the union over n gives $\mathbb{P}(Y' > Y,\ Y < \infty) = 0$. Since $Y' > Y$ forces $Y < \infty$, we get $\mathbb{P}(Y' > Y) = 0$. By symmetry $\mathbb{P}(Y > Y') = 0$.

5. Take $\Omega = \{a, b, c\}$, $\mathcal{F} = 2^\Omega$, $\mathbb{P}$ uniform, $X = \mathbf{1}_{\{a\}}$, $\mathcal{G} = \sigma(\{a\})$, $\mathcal{H} = \sigma(\{c\})$. Then $\mathbb{E}[X|\mathcal{G}] = X$ and

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}] = \tfrac{1}{2}\mathbf{1}_{\{a,b\}}, \qquad \mathbb{E}[X|\mathcal{H}] = \tfrac{1}{2}\mathbf{1}_{\{a,b\}}, \qquad \mathbb{E}[\mathbb{E}[X|\mathcal{H}]|\mathcal{G}] = \tfrac{1}{2}\mathbf{1}_{\{a\}} + \tfrac{1}{4}\mathbf{1}_{\{b,c\}}.$$

At c (which has probability $\frac{1}{3}$) the first equals 0 and the second equals $\frac{1}{4}$.

Martingales

1. ($\Rightarrow$) For $A \in \mathcal{F}_s$, $\mathbb{E}[X_t\mathbf{1}_A] = \mathbb{E}[\mathbb{E}[X_t|\mathcal{F}_s]\mathbf{1}_A] \ge \mathbb{E}[X_s\mathbf{1}_A]$. ($\Leftarrow$) Let $Z = \mathbb{E}[X_t|\mathcal{F}_s] - X_s$, which is $\mathcal{F}_s$-measurable with $\mathbb{E}[Z\mathbf{1}_A] \ge 0$ for all $A \in \mathcal{F}_s$. With $A = \{Z < 0\}$ we get $\mathbb{E}[Z\mathbf{1}_{\{Z<0\}}] \ge 0$, so $\mathbb{P}(Z < 0) = 0$, i.e. $\mathbb{E}[X_t|\mathcal{F}_s] \ge X_s$.

2. (The index set in $(X_t, \mathcal{F}_t^X)_{t\in X}$ should be T.) ($\Leftarrow$) Take $\mathcal{F}_t = \mathcal{F}_t^X$. ($\Rightarrow$) Since X is $(\mathcal{F}_t)$-adapted, $\mathcal{F}_t^X \subseteq \mathcal{F}_t$, and X is $\mathcal{F}^X$-adapted and integrable. For $s < t$, by the tower property,

$$\mathbb{E}[X_t|\mathcal{F}_s^X] = \mathbb{E}[\mathbb{E}[X_t|\mathcal{F}_s]|\mathcal{F}_s^X] = \mathbb{E}[X_s|\mathcal{F}_s^X] = X_s.$$

3. (Doob's theorem; (a) should read $\mathbb{E}|X_{\tau_2}| < \infty$.) Here $\mathcal{F}_\infty \supseteq \mathcal{F}_n$ for all n, and the martingale property gives $X_n = \mathbb{E}[X_\infty|\mathcal{F}_n]$. A stopping time is $\tau : \Omega \to \mathbb{N}_\infty$ with $\{\tau = n\} \in \mathcal{F}_n$, $n \in \mathbb{N}$, and $\mathcal{F}_\tau = \{A \in \mathcal{F}_\infty : A \cap \{\tau = n\} \in \mathcal{F}_n\ \forall n \in \mathbb{N}\}$.

Claim: for every stopping time τ, $X_\tau = \mathbb{E}[X_\infty|\mathcal{F}_\tau]$.

$X_\tau = \sum_{n\in\mathbb{N}_\infty} X_n\mathbf{1}_{\{\tau=n\}}$ is $\mathcal{F}_\tau$-measurable, since $\{X_\tau \in B\} \cap \{\tau = n\} = \{X_n \in B\} \cap \{\tau = n\} \in \mathcal{F}_n$. By conditional Jensen, $|X_n| \le \mathbb{E}[|X_\infty||\mathcal{F}_n]$, so

$$\mathbb{E}|X_\tau| \le \sum_n \mathbb{E}[|X_\infty|\mathbf{1}_{\{\tau=n\}}] = \mathbb{E}|X_\infty|.$$

For $A \in \mathcal{F}_\tau$, Fubini (justified by the last bound) gives

$$\mathbb{E}[X_\tau\mathbf{1}_A] = \sum_n \mathbb{E}[X_n\mathbf{1}_{A\cap\{\tau=n\}}] = \sum_n \mathbb{E}[X_\infty\mathbf{1}_{A\cap\{\tau=n\}}] = \mathbb{E}[X_\infty\mathbf{1}_A],$$

because $A \cap \{\tau = n\} \in \mathcal{F}_n$ (trivially for $n = \infty$).

(a) $\mathbb{E}|X_{\tau_2}| \le \mathbb{E}|X_\infty| < \infty$.

(b) If $A \in \mathcal{F}_{\tau_1}$ then $A \cap \{\tau_2 \le n\} = A \cap \{\tau_1 \le n\} \cap \{\tau_2 \le n\} \in \mathcal{F}_n$, so $\mathcal{F}_{\tau_1} \subseteq \mathcal{F}_{\tau_2}$. By the claim and the tower property,

$$\mathbb{E}[X_{\tau_2}|\mathcal{F}_{\tau_1}] = \mathbb{E}[\mathbb{E}[X_\infty|\mathcal{F}_{\tau_2}]|\mathcal{F}_{\tau_1}] = \mathbb{E}[X_\infty|\mathcal{F}_{\tau_1}] = X_{\tau_1}.$$

4. $\mathcal{F}_\tau = \{A : A \cap \{\tau \le u\} \in \mathcal{F}_u \ \forall u\}$. From discrete time we use: if $(Z_g, \mathcal{F}_g)_{g\in G}$ is a supermartingale on a discrete set $G \subset [0,\infty)$ and $\sigma \le \rho \le c$ are G-valued stopping times, then $\mathbb{E}[Z_\rho|\mathcal{F}_\sigma] \le Z_\sigma$, and the stopped process $(Z_{g\wedge\rho})_{g\in G}$ is a supermartingale.

Lemma. Let Y_n be integrable and $\mathcal{H}_n$-measurable with $\mathbb{E}[Y_m\mathbf{1}_B] \le \mathbb{E}[Y_n\mathbf{1}_B]$ for all $m \le n$ and $B \in \mathcal{H}_n$, and suppose $L = \sup_n \mathbb{E}Y_n < \infty$. Then (Y_n) is uniformly integrable.

Proof. Taking $B = \Omega$ shows $\mathbb{E}Y_n \uparrow L$. From $Y_n \ge \mathbb{E}[Y_1|\mathcal{H}_n] \ge -\mathbb{E}[Y_1^-|\mathcal{H}_n]$ we get $\mathbb{E}Y_n^- \le \mathbb{E}Y_1^-$, so $\sup_n \mathbb{E}|Y_n| \le L + 2\mathbb{E}|Y_1| =: K$. Given $\varepsilon > 0$ pick m with $\mathbb{E}Y_m \ge L - \varepsilon$. For $n \ge m$, since $\{Y_n \le \lambda\}, \{Y_n < -\lambda\} \in \mathcal{H}_n$,

$$\begin{aligned}\mathbb{E}[|Y_n|; |Y_n| > \lambda] &= \mathbb{E}Y_n - \mathbb{E}[Y_n; Y_n \le \lambda] - \mathbb{E}[Y_n; Y_n < -\lambda] \\ &\le \mathbb{E}Y_m + \varepsilon - \mathbb{E}[Y_m; Y_n \le \lambda] - \mathbb{E}[Y_m; Y_n < -\lambda] \le \mathbb{E}[|Y_m|; |Y_n| > \lambda] + \varepsilon.\end{aligned}$$

As $\mathbb{P}(|Y_n| > \lambda) \le K/\lambda$ and Y_m is integrable, the right side is $\le 2\varepsilon$ for λ large, uniformly in $n \ge m$. The finitely many $n < m$ are handled trivially. □

(a) Fix $s < t$ and let $G_n = 2^{-n}\mathbb{N} \cup \{s,t\}$; these sets are nested. Let $\tau_n = \inf\{g \in G_n : g > \tau\}$, with $\tau_n = \infty$ if $\tau = \infty$. Then $\{\tau_n \le g\} = \{\tau < g\} \in \mathcal{F}_g$, so τ_n is a G_n-valued stopping time, and it is also a stopping time for the grid $G_{n'}$, $n' \ge n$. Moreover $\tau < \tau_n \le \tau + 2^{-n}$ and $\tau_n \downarrow \tau$.

For $r \in \{s,t\}$, $X_{r\wedge\tau_n} = \sum_{g\in G_n\cap[0,r]} X_g\mathbf{1}_{\{r\wedge\tau_n = g\}}$ is $\mathcal{F}_r$-measurable. By right-continuity $X_{r\wedge\tau_n} \to X_{r\wedge\tau}$, so $X_{r\wedge\tau}$ is $\mathcal{F}_r$-measurable. For $u < r$, $\{X_{r\wedge\tau} \in B\} \cap \{\tau \le u\} = \{X_{u\wedge\tau} \in B\} \cap \{\tau \le u\} \in \mathcal{F}_u$, hence $X_{r\wedge\tau}$ is $\mathcal{F}_{r\wedge\tau}$-measurable.

Since the stopped discrete process is a supermartingale and $s, t \in G_n$, for $A \in \mathcal{F}_s$,

$$\mathbb{E}[X_{t\wedge\tau_n}\mathbf{1}_A] \le \mathbb{E}[X_{s\wedge\tau_n}\mathbf{1}_A]. \tag{$*$}$$

Let $Y_n = X_{r\wedge\tau_n}$ and let $\mathcal{H}_n$ be the σ-algebra of $r \wedge \tau_n$ on the grid G_n. For $m \le n$, $r\wedge\tau_n \le r\wedge\tau_m \le r$ are G_n-stopping times, so $\mathbb{E}[Y_m|\mathcal{H}_n] \le Y_n$. Also $\mathbb{E}Y_n \le \mathbb{E}X_0$. By the Lemma, (Y_n) is UI. Together with a.s. convergence this gives $X_{r\wedge\tau_n} \to X_{r\wedge\tau}$ in L^1, so $X_{r\wedge\tau}$ is integrable. Passing to the limit in $(*)$ gives $\mathbb{E}[X_{t\wedge\tau}\mathbf{1}_A] \le \mathbb{E}[X_{s\wedge\tau}\mathbf{1}_A]$ for $A \in \mathcal{F}_s$. By Problem 1 (with reversed inequality), $(X_{t\wedge\tau}, \mathcal{F}_t)$ is a supermartingale.

Since $\mathcal{F}_{s\wedge\tau} \subseteq \mathcal{F}_s$ (if $A \in \mathcal{F}_{s\wedge\tau}$ then $A = A \cap \{s\wedge\tau \le s\} \in \mathcal{F}_s$), $(\mathcal{F}_{t\wedge\tau})$ is a filtration to which $X_{t\wedge\tau}$ is adapted. The same inequality for $A \in \mathcal{F}_{s\wedge\tau}$ shows that $(X_{t\wedge\tau}, \mathcal{F}_{t\wedge\tau})$ is a supermartingale.

(b) Applying (a) to X and $-X$, $(X_{t\wedge\tau})$ is a martingale, so $\mathbb{E}X_{t\wedge\tau} = \mathbb{E}X_0$. On $\{\tau < \infty\}$, $X_{t\wedge\tau} = X_\tau$ for $t \ge \tau$. On $\{\tau = \infty\}$, X_τ must be given a meaning. We set $X_\tau := \lim_t X_{t\wedge\tau}$, which exists a.s. by Problem 5(a) because $\sup_t \mathbb{E}X_{t\wedge\tau}^+ \le \mathbb{E}Y$. (If $\tau < \infty$ a.s. this is not needed.) Then $|X_\tau| \le Y$, so X_τ is integrable, and dominated convergence gives $\mathbb{E}X_\tau = \lim_t \mathbb{E}X_{t\wedge\tau} = \mathbb{E}X_0$.

5.

(a) Let D be the non-negative dyadic rationals and $F_k = D_k \cap [0,k]$. For $a < b$ let U_k be the number of upcrossings of $[a,b]$ by $(X_t)_{t\in F_k}$. By Doob's upcrossing inequality for the discrete submartingale on the finite set F_k,

$$(b-a)\mathbb{E}U_k \le \mathbb{E}(X_k - a)^+ \le \sup_t \mathbb{E}X_t^+ + |a|.$$

U_k increases to the number U_D of upcrossings by $(X_t)_{t\in D}$, so by monotone convergence $\mathbb{E}U_D < \infty$. Hence a.s. $U_D[a,b] < \infty$ for all rational $a < b$, and $\lim_{D\ni t\to\infty} X_t$ exists in $[-\infty,\infty]$ (otherwise $\liminf < a < b < \limsup$ would give infinitely many upcrossings).

By right-continuity, if $X_t > c$ for arbitrarily large real t then $X_d > c$ for arbitrarily large $d \in D$. So $\limsup_{t\to\infty} X_t = \limsup_{D\ni t\to\infty} X_t$, and similarly for $\liminf$. Thus $X = \lim_{t\to\infty} X_t$ exists a.s. Since $\mathbb{E}|X_t| = 2\mathbb{E}X_t^+ - \mathbb{E}X_t \le 2\sup_t \mathbb{E}X_t^+ - \mathbb{E}X_0$, Fatou's lemma gives $\mathbb{E}|X| < \infty$.

(b)

Remark. Right-continuity must be assumed here as well. Let $U \sim U[0,1]$ and $X_t = \mathbf{1}_{\{t-\lfloor t \rfloor = U\}}$. Then $X_t = 0$ a.s. for each t, so X is an L^2-bounded martingale, but every path equals 1 at $U + k$ for all $k \in \mathbb{N}$ and does not converge.

Assume right-continuity. Since $\sup \mathbb{E}X_t^+ \le \sup(\mathbb{E}X_t^2)^{1/2} < \infty$, (a) gives $X_t \to X$ a.s. For $s < t$, by Conditional expectation, Problem 2(b), $\mathbb{E}[X_s(X_t - X_s)] = \mathbb{E}[X_s \mathbb{E}[X_t - X_s|\mathcal{F}_s]] = 0$. Hence $\mathbb{E}(X_t - X_s)^2 = \mathbb{E}X_t^2 - \mathbb{E}X_s^2$. So $\mathbb{E}X_t^2$ is non-decreasing and bounded, and $\mathbb{E}(X_t - X_s)^2 \to 0$ as $s, t \to \infty$. By completeness of L^2, $X_t \to X'$ in L^2. Along a sequence $t_n \to \infty$ a subsequence converges a.s. to X', so $X' = X$ a.s. and $X \in L^2$. Finally, for $u > t$, $X_t = \mathbb{E}[X_u|\mathcal{F}_t]$ and

$$\|\mathbb{E}[X_u|\mathcal{F}_t] - \mathbb{E}[X|\mathcal{F}_t]\|_1 \le \|X_u - X\|_1 \to 0,$$

so $\mathbb{E}[X|\mathcal{F}_t] = X_t$ a.s.

6.

(a) The argument works for every $\sigma \in \mathbb{R}$, which we need in (b). Let $M_t(\sigma) = \exp(\sigma B_t - \sigma^2 t/2)$. It is adapted, and integrable since $\mathbb{E}e^{\sigma B_t} = e^{\sigma^2 t/2}$. Since $B_t - B_s$ is independent of $\mathcal{F}_s$,

$$\mathbb{E}[M_t|\mathcal{F}_s] = M_s \, \mathbb{E}\big[e^{\sigma(B_t - B_s)}\big] e^{-\sigma^2(t-s)/2} = M_s.$$

(b) Writing $x = B_t$, $e^{\sigma x - \sigma^2 t/2} = e^{\sigma x} e^{-\sigma^2 t/2} = \sum_k \frac{\sigma^k}{k!} H_k(x,t)$, where

$$H_2 = x^2 - t, \quad H_3 = x^3 - 3tx, \quad H_4 = x^4 - 6tx^2 + 3t^2$$

are read off from the coefficients of $\sigma^2, \sigma^3, \sigma^4$ in $(1 + \sigma x + \frac{\sigma^2 x^2}{2} + \frac{\sigma^3 x^3}{6} + \frac{\sigma^4 x^4}{24})(1 - \frac{\sigma^2 t}{2} + \frac{\sigma^4 t^2}{8})$. Thus $\partial_\sigma^k M_t(\sigma)|_{\sigma=0} = H_k(B_t, t)$.

By (a), $\mathbb{E}[M_t(\sigma)\mathbf{1}_A] = \mathbb{E}[M_s(\sigma)\mathbf{1}_A]$ for $A \in \mathcal{F}_s$ and all σ. For $|\sigma| \le 1$, $\partial_\sigma^j M_t(\sigma) = P_j(B_t - \sigma t, t) M_t(\sigma)$ with P_j a polynomial. This is dominated by $C_j(1 + |B_t| + t)^j e^{|B_t|}$, which is integrable, so we may differentiate under $\mathbb{E}$ k times at $\sigma = 0$: $\mathbb{E}[H_k(B_t,t)\mathbf{1}_A] = \mathbb{E}[H_k(B_s,s)\mathbf{1}_A]$. Since $H_k(B_t,t)$ is adapted and integrable, it is a martingale. For $k = 2, 3, 4$ this gives the processes $B_t^2 - t$, $B_t^3 - 3tB_t$ and $B_t^4 - 6tB_t^2 + 3t^2$.

Exercise 4

Stochastic calculus

Throughout, for $f \in L^2$ (adapted, with $\mathbb{E}\int_0^t f^2 ds < \infty$), $\int_0^t f\, dB$ is the L^2-limit of $\int f_n\, dB$ for simple adapted $f_n \to f$ in $L^2(ds \otimes d\mathbb{P})$. For a partition $0 = t_0 < \cdots < t_n = t$ of mesh $|\pi|$ write $\Delta_i = B_{t_i} - B_{t_{i-1}}$ and $\Delta t_i = t_i - t_{i-1}$.

1.

Remark. In (a)ii and (b)i the right-hand side should be $\frac{1}{3}B_t^3 - \int_0^t B_s\, ds$, not $\frac{1}{3}B_t^2 - \dots$. As stated it cannot hold: the left side has mean 0, while $\mathbb{E}[\frac{1}{3}B_t^2 - \int_0^t B_s\, ds] = \frac{t}{3}$. Itô's formula in (b) confirms the corrected form.

(a) i. $f_\pi = \sum t_{i-1}\mathbf{1}_{[t_{i-1},t_i)} \to s$ in L^2, since $\int_0^t |f_\pi(s) - s|^2 ds \le t|\pi|^2$. Summation by parts gives

$$\sum_i t_{i-1}\Delta_i = tB_t - \sum_i B_{t_i}\Delta t_i.$$

By continuity $|\sum_i B_{t_i}\Delta t_i - \int_0^t B_s\, ds| \to 0$ pathwise. It is bounded by $2t \sup_{[0,t]}|B| \in L^2$, so the convergence also holds in L^2. Hence $\int_0^t s\, dB_s = tB_t - \int_0^t B_s\, ds$.

(a) ii. For $u < s \le t$, by Cauchy–Schwarz and $B_s+B_u \sim N(0, s+3u)$, $\mathbb{E}(B_s^2-B_u^2)^2 \le \sqrt{3(s-u)^2}\sqrt{3(4t)^2} = 12t(s-u)$. Hence $\sum_i B_{t_{i-1}}^2 \mathbf{1}_{[t_{i-1},t_i)} \to B^2$ in $L^2(ds \otimes d\mathbb{P})$, and $\int_0^t B^2 dB = \lim_{L^2} S_\pi$ with $S_\pi = \sum_i B_{t_{i-1}}^2 \Delta_i$. Summing $b^3 - a^3 = 3a^2(b-a) + 3a(b-a)^2 + (b-a)^3$ over the partition gives

$$S_\pi = \tfrac{1}{3}B_t^3 - \sum_i B_{t_{i-1}}\Delta_i^2 - \tfrac{1}{3}\sum_i \Delta_i^3.$$

The cross terms below vanish by independence of increments and since odd moments are 0, resp. $\mathbb{E}[\Delta_j^2 - \Delta t_j] = 0$:

$$\mathbb{E}\Big(\sum_i \Delta_i^3\Big)^2 = \sum_i 15\Delta t_i^3 \le 15t|\pi|^2, \qquad \mathbb{E}\Big(\sum_i B_{t_{i-1}}(\Delta_i^2 - \Delta t_i)\Big)^2 = \sum_i t_{i-1}\cdot 2\Delta t_i^2 \le 2t^2|\pi|.$$

Moreover $\sum_i B_{t_{i-1}}\Delta t_i \to \int_0^t B_s\,ds$ in L^2 as in i. Hence $\int_0^t B_s^2\,dB_s = \frac{1}{3}B_t^3 - \int_0^t B_s\,ds$.

(b) i. Itô's formula with $f(x) = x^3/3$ (so $f' = x^2$, $f'' = 2x$) gives $\frac{1}{3}B_t^3 = \int_0^t B_s^2\,dB_s + \int_0^t B_s\,ds$.

2.

Remark. We take $f, g \in L^2([0,T]\times\Omega)$, i.e. $\mathbb{E}\int_0^T f^2 < \infty$ for every T, the class on which the Itô integral is an L^2-martingale. If only $\int_0^T f^2 ds < \infty$ a.s. were assumed, the claim is false in general; we do not pursue this.

The Itô integral has mean zero, so taking expectations gives $C = D$. Then $\int_0^T (f-g)\,dB = 0$ a.s., and by the Itô isometry

$$\mathbb{E}\int_0^T (f-g)^2 ds = \mathbb{E}\Big(\int_0^T (f-g)\,dB\Big)^2 = 0,$$

i.e. $f = g$ for $ds \otimes d\mathbb{P}$-a.e. (s,ω).

3.

(a) $d(B_t^2) = dt + 2B_t\,dB_t$.

(b) $d(2 + t + e^{B_t}) = (1 + \frac{1}{2}e^{B_t})\,dt + e^{B_t}\,dB_t$.

(c) $d(B_t^2 + \tilde{B}_t^2) = 2\,dt + 2B_t\,dB_t + 2\tilde{B}_t\,d\tilde{B}_t$, i.e. $\alpha = 2$ and $\beta = (2B_t, 2\tilde{B}_t)$ against $d(B_t, \tilde{B}_t)$.

(d) $d(t, B_t) = \binom{1}{0}dt + \binom{0}{1}dB_t$.

(e) With $f(t,x) = e^{t/2}\cos x$: $f_t = \frac{1}{2}f$, $f_x = -e^{t/2}\sin x$, $f_{xx} = -f$. Hence

$$d\big(e^{t/2}\cos B_t\big) = \big(\tfrac{1}{2}f - \tfrac{1}{2}f\big)dt - e^{t/2}\sin B_t\,dB_t = -e^{t/2}\sin B_t\,dB_t.$$

So $e^{t/2}\cos B_t = 1 - \int_0^t e^{s/2}\sin B_s\,dB_s$. The integrand is bounded by $e^{T/2}$ on $[0,T]$, so it lies in L^2 and the Itô integral is a martingale. Therefore $e^{t/2}\cos B_t$ is a martingale.

4.

(a) With $f = e^x$, $f' = f'' = f$: $dX_t = e^{B_t}dB_t + \frac{1}{2}e^{B_t}dt = \frac{1}{2}X_t dt + X_t dB_t$.

(b) With $f = \sin x$: $dX_t = \cos B_t\,dB_t - \frac{1}{2}\sin B_t\,dt$. While $\cos B_t > 0$, i.e. for $t < \inf\{s : |B_s| = \pi/2\}$, we have $\cos B_t = \sqrt{1-\sin^2 B_t} = \sqrt{1-X_t^2}$. Hence $dX_t = -\frac{1}{2}X_t dt + \sqrt{1-X_t^2}\,dB_t$.

5. For $k \ge 2$, Itô's formula gives

$$B_t^k = \int_0^t kB_s^{k-1}dB_s + \tfrac{1}{2}k(k-1)\int_0^t B_s^{k-2}ds.$$

The integrand satisfies $\mathbb{E}\int_0^t B_s^{2k-2}ds < \infty$, so the stochastic integral has mean 0. By Fubini,

$$\beta_k(t) = \tfrac{1}{2}k(k-1)\int_0^t \beta_{k-2}(s)\,ds.$$

The formula also holds for $k = 1$ ($\beta_1 = 0$) but fails for $k = 0$, where $\beta_0 = 1$. With $\beta_0 = 1$: $\beta_2(t) = t$, then $\mathbb{E}B_t^4 = \beta_4(t) = 6\int_0^t s\,ds = 3t^2$ and $\mathbb{E}B_t^6 = \beta_6(t) = 15\int_0^t 3s^2\,ds = 15t^3$.

6.

Remark. g_ε is not C^2 on $\mathbb{R}$. It is C^1, with $g_\varepsilon'(x) = \operatorname{sgn} x$ for $|x| \geq \varepsilon$ and x/ε for $|x| < \varepsilon$. Its second derivative $g_\varepsilon'' = \varepsilon^{-1}\mathbf{1}_{(-\varepsilon,\varepsilon)}$ exists except at $\pm\varepsilon$, so g_ε is C^2 on $\mathbb{R}\setminus\{\pm\varepsilon\}$. Itô's formula still applies, as shown in (a).

(a) Let ϕ_n be continuous, $0 \leq \phi_n \leq 1/\varepsilon$, with $\phi_n = g_\varepsilon''$ outside $(\pm\varepsilon - \frac{1}{n}, \pm\varepsilon + \frac{1}{n})$. Let $h_n(x) = g_\varepsilon(0) + \int_0^x\int_0^y \phi_n$, which is C^2. Since $g_\varepsilon'(0) = 0$, $|h_n' - g_\varepsilon'| \leq \int|\phi_n - g_\varepsilon''| \leq 4/(n\varepsilon)$ and $|h_n - g_\varepsilon|(x) \leq 4|x|/(n\varepsilon)$. Itô's formula for h_n reads

$$h_n(B_t) = h_n(B_0) + \int_0^t h_n'(B_s)\,dB_s + \tfrac{1}{2}\int_0^t \phi_n(B_s)\,ds.$$

As $n \to \infty$: $h_n(B_t) \to g_\varepsilon(B_t)$, and $\int h_n'(B)dB \to \int g_\varepsilon'(B)dB$ in L^2 by the isometry and uniform convergence. Also $\phi_n(B_s) \to g_\varepsilon''(B_s)$ for all s with $B_s \neq \pm\varepsilon$, which is a.e. s a.s. because $\mathbb{E}\operatorname{Leb}\{s \leq t : B_s = \pm\varepsilon\} = \int_0^t \mathbb{P}(B_s = \pm\varepsilon)ds = 0$. Bounded convergence then gives

$$g_\varepsilon(B_t) = g_\varepsilon(B_0) + \int_0^t g_\varepsilon'(B_s)\,dB_s + \frac{1}{2\varepsilon}\operatorname{Leb}\{s \in [0,t] : B_s \in (-\varepsilon,\varepsilon)\}.$$

(b) By the isometry and dominated convergence ($\mathbb{P}(B_s = 0) = 0$ for $s > 0$),

$$\mathbb{E}\Big(\int_0^t \tfrac{B_s}{\varepsilon}\mathbf{1}_{\{|B_s|<\varepsilon\}}dB_s\Big)^2 = \mathbb{E}\int_0^t \tfrac{B_s^2}{\varepsilon^2}\mathbf{1}_{\{|B_s|<\varepsilon\}}ds \leq \int_0^t \mathbb{P}(|B_s| < \varepsilon)\,ds \xrightarrow[\varepsilon\to 0]{} 0.$$

(c) Split $\int g_\varepsilon'(B)dB = \int \operatorname{sgn}(B)\mathbf{1}_{\{|B|\geq\varepsilon\}}dB + \int \frac{B}{\varepsilon}\mathbf{1}_{\{|B|<\varepsilon\}}dB$. The second term tends to 0 in L^2 by (b). The first tends to $\int_0^t \operatorname{sgn}(B_s)dB_s$ in L^2, since $\mathbb{E}\int_0^t \mathbf{1}_{\{|B_s|<\varepsilon\}}ds \to 0$. Moreover $0 \leq g_\varepsilon(x) - |x| = (\varepsilon - |x|)^2/(2\varepsilon)\,\mathbf{1}_{\{|x|<\varepsilon\}} \leq \varepsilon/2$. Hence by (a)

$$\frac{1}{2\varepsilon}\operatorname{Leb}\{s \leq t : B_s \in (-\varepsilon,\varepsilon)\} = g_\varepsilon(B_t) - g_\varepsilon(B_0) - \int_0^t g_\varepsilon'(B_s)dB_s$$

converges in L^2 to $|B_t| - |B_0| - \int_0^t \operatorname{sgn}(B_s)dB_s$. Thus L_t exists as an L^2-limit and $|B_t| = |B_0| + \int_0^t \operatorname{sgn}(B_s)dB_s + L_t$.

7.

Remark. The definition should read $T_x = \inf\{t > 0 : X_t^x = 0\}$ with $x > 0$, not $\inf\{t : X_t^x > 0\}$. In (b) the claim is $(Y_t) \overset{d}{=} (X_t^1)$. All X^x, $x > 0$, are driven by the same B.

Facts used: X^x is continuous and positive on $[0, T_x)$; $B_t/t \to 0$ a.s.; and the following consequence of the equation. If $X_t^x < R$ for all $t \geq 0$, then $T_x = \infty$ and $X_t^x \geq x + B_t + at/R \to \infty$, a contradiction. So X^x cannot stay in (r, R) forever.

(a) For $t < S = T_x \wedge T_y$, $D_t = X_t^y - X_t^x = y - x + a\int_0^t \big(\frac{1}{X^y} - \frac{1}{X^x}\big)ds$ is C^1 with $D' = -aD/(X^xX^y)$. Hence

$$D_t = (y-x)\exp\Big(-a\int_0^t \frac{ds}{X_s^xX_s^y}\Big) > 0.$$

If $T_y < T_x$, then $0 < X_t^x < X_t^y \to 0$ as $t \uparrow T_y$, so $X^x_{T_y} = 0$, a contradiction. So $T_x \le T_y$ and $X_t^x < X_t^y$ for $t < T_x$. With $y = x$, the same computation gives $D \equiv 0$: the equation has at most one solution for a given path of B (pathwise uniqueness).

(b) Let $W_t = x^{-1} B_{x^2 t}$, a BM. Substituting $s = x^2 u$ in $\int_0^{x^2 t} ds/X_s^x = x^2 \int_0^t du/X^x_{x^2 u} = x \int_0^t du/Y_u$,

$$Y_t = 1 + W_t + a \int_0^t \frac{du}{Y_u}, \qquad t < T_x/x^2.$$

So Y solves the equation with $x = 1$ driven by W, and its lifetime is T_x/x^2. By pathwise uniqueness the solution is a fixed functional of the driving path, $X^1 = \Phi(B)$, and therefore $Y = \Phi(W) \overset{d}{=} \Phi(B) = X^1$.

(c) Let $s(x) = x^{1-2a}$ for $a \neq \frac{1}{2}$ and $s(x) = \log x$ for $a = \frac{1}{2}$. Then $\frac{1}{2}s'' + \frac{a}{x}s' = 0$. For $0 < r < x < R$ let τ be the exit time of X^x from (r, R); it is a.s. finite by the facts above. Itô's formula for $dX = \frac{a}{X}dt + dB$ gives $s(X_{t\wedge\tau}) = s(x) + \int_0^{t\wedge\tau} s'(X) dB$, whose integrand is bounded, so the integral is a martingale. Bounded convergence yields $\mathbb{E}s(X_\tau) = s(x)$, i.e.

$$p_{r,R}(x) := \mathbb{P}(X^x \text{ hits } r \text{ before } R) = \frac{s(R) - s(x)}{s(R) - s(r)}.$$

By (a), $T_x \le T_y$ and $X^x \le X^y$ for $x < y$. Hence it suffices to prove i, ii and iv for the countably many $x \in \mathbb{N}$, and iii for $x = 1/n$, $n \in \mathbb{N}$.

i. ($a \ge \frac{1}{2}$.) If $T_x < \infty$, then $\sup_{[0,T_x]} X^x < R$ for some R, so X^x hits every r before R. But $p_{r,R}(x) \to 0$ as $r \to 0$, since $s(r) \to -\infty$ for $a = \frac{1}{2}$, and for $a > \frac{1}{2}$ we have $s(r) \to +\infty$ while $s(R) - s(x)$ is finite. Hence $\mathbb{P}(T_x < \infty) \le \sum_R \lim_{r\to 0} p_{r,R}(x) = 0$.

ii. ($a = \frac{1}{2}$.) By i, $T_x = \infty$. $\mathbb{P}(X^x \text{ hits } r) \ge p_{r,R}(x) = \frac{\log R - \log x}{\log R - \log r} \to 1$ as $R \to \infty$. Intersecting over $r = 1/n$ gives $\inf_t X_t^x = 0$ a.s.

iii. ($a > \frac{1}{2}$.) Here $T_x = \infty$ and $M_t = (X_t^x)^{1-2a} > 0$. With $\sigma_n = \inf\{t : X_t^x \notin (\frac{1}{n}, n)\} \uparrow \infty$, the process $M_{t\wedge\sigma_n}$ is a bounded martingale. Conditional Fatou gives $\mathbb{E}[M_t|\mathcal{F}_s] \le \liminf_n M_{s\wedge\sigma_n} = M_s$, so M is a non-negative continuous supermartingale. By Exercise 2, Martingales 5(a) applied to $-M$, $M_t \to M_\infty \in [0, \infty)$ a.s., hence $X_t^x \to X_\infty \in (0, \infty]$. If $X_\infty < \infty$ then X^x is bounded, which contradicts the facts above. So $X_t^x \to \infty$.

iv. ($a < \frac{1}{2}$.) Here $s(0) = 0$ and $s(\infty) = \infty$. The events $E_r = \{X^x \text{ hits } r \text{ before } R\}$ decrease as $r \downarrow 0$. On $\bigcap_r E_r$ the hitting times τ_r increase to some $\sigma \le \tau_R$. Moreover $\sigma < \infty$ a.s.: $\tau_R < \infty$ when $T_x = \infty$ by the facts above, and $\sigma \le T_x$ otherwise. By continuity $X^x_\sigma = 0$, so $\sigma = T_x < \tau_R$. Hence

$$\mathbb{P}(T_x < \tau_R) = \lim_{r\to 0} p_{r,R}(x) = 1 - (x/R)^{1-2a} \to 1 \quad (R \to \infty),$$

i.e. $T_x < \infty$ a.s.

For v and vi fix $y > x > 0$, write $X = X^x$, $D = X^y - X^x$, and let $V = X/D$ on $[0, T_x)$. D is C^1, $D > 0$ on $[0, T_x)$, and $T_x < \infty$ a.s. by iv. By (a), $T_x = T_y$ iff $D_{T_x} = 0$. From $d(1/D) = \frac{a}{XX^y D}dt$, $X = DV$ and $X^y = D(V + 1)$,

$$dV = \frac{1}{D}\Big(\frac{a}{X}dt + dB\Big) + X\, d\Big(\frac{1}{D}\Big) = \frac{1}{D^2}\Big(\frac{a}{V} + \frac{a}{V+1}\Big)dt + \frac{1}{D}\, dB.$$

For $\varphi \in C^2(0, \infty)$, Itô's formula gives $d\varphi(V) = D^{-2}\big(\frac{1}{2}\varphi'' + (\frac{a}{V} + \frac{a}{V+1})\varphi'\big)(V)\, dt + \varphi'(V)D^{-1}dB$. Let

$$S(v) = \int_0^v \big(u(1+u)\big)^{-2a} du,$$

which is finite since $a < \frac{1}{2}$. Then $\frac{1}{2}S'' + (\frac{a}{v} + \frac{a}{v+1})S' = 0$, and $S(V_t)$ has no dt term. Let $\rho_n = \inf\{t < T_x : D_t \le \frac{1}{n} \text{ or } V_t \le \frac{1}{n}\} \wedge n \wedge T_x$. The integrand of $S(V)$ is bounded up to ρ_n, so $\mathbb{E}S(V_{\rho_n\wedge\kappa}) = S(V_0)$ for

every stopping time κ. Moreover $\rho_n \uparrow T_x$: if the limit were $< T_x$, then D and V would be bounded away from 0 on a compact interval past it.

v. ($\frac{1}{4} < a < \frac{1}{2}$.) Now $S(\infty) < \infty$, since $(u(1+u))^{-2a} \sim u^{-4a}$ and $4a > 1$. Suppose $\mathbb{P}(T_x < T_y) = 1$. On $\{T_x < T_y\}$ we have $D_{T_x} > 0$ and $X_t \to 0$, so $V_t \to 0$ as $t \uparrow T_x$. Hence for large n, ρ_n is the hitting time of $\frac{1}{n}$ by V and $S(V_{\rho_n}) = S(\frac{1}{n}) \to 0$. Since $0 \le S \le S(\infty)$, bounded convergence gives $S(V_0) = \lim_n \mathbb{E}S(V_{\rho_n}) = 0$, contradicting $V_0 = x/(y-x) > 0$. Hence $\mathbb{P}(T_x = T_y) > 0$.

vi. ($a \le \frac{1}{4}$.) Now $S(\infty) = \infty$. Let $\kappa_\lambda = \inf\{t : S(V_t) \ge \lambda\}$. Since $S \ge 0$, $\lambda\mathbb{P}(\kappa_\lambda \le \rho_n) \le \mathbb{E}S(V_{\rho_n \wedge \kappa_\lambda}) = S(V_0)$. Letting $n \to \infty$ and then $\lambda \to \infty$ gives $\sup_{t<T_x} V_t < \infty$ a.s. Now fix K and let $\rho_n' = \inf\{t : D_t \le \frac{1}{n} \text{ or } V_t \ge K\} \wedge n \wedge T_x$. Taking expectations in the equation for V (with $d\sigma = dt/D^2$, and the stochastic integral having bounded integrand $1/D \le n$),

$$\mathbb{E}\int_0^{\rho_n'} \frac{a}{V_t}\frac{dt}{D_t^2} \le \mathbb{E}V_{\rho_n'} \le K.$$

On $\{\sup_{t<T_x} V < K\}$ we have $\rho_n' \uparrow T_x$, so monotone convergence gives $\int_0^{T_x} \frac{dt}{V_t D_t^2} < \infty$ a.s. there. Since $\frac{1}{XX^y} = \frac{1}{D^2V(V+1)} \le \frac{1}{D^2V}$,

$$\log D_{T_x} = \log(y-x) - a\int_0^{T_x} \frac{dt}{X_t X_t^y} > -\infty,$$

i.e. $D_{T_x} > 0$ and $T_x < T_y$. Letting $K \to \infty$ along $\mathbb{N}$, $T_x < T_y$ a.s.